Video Motion Detection

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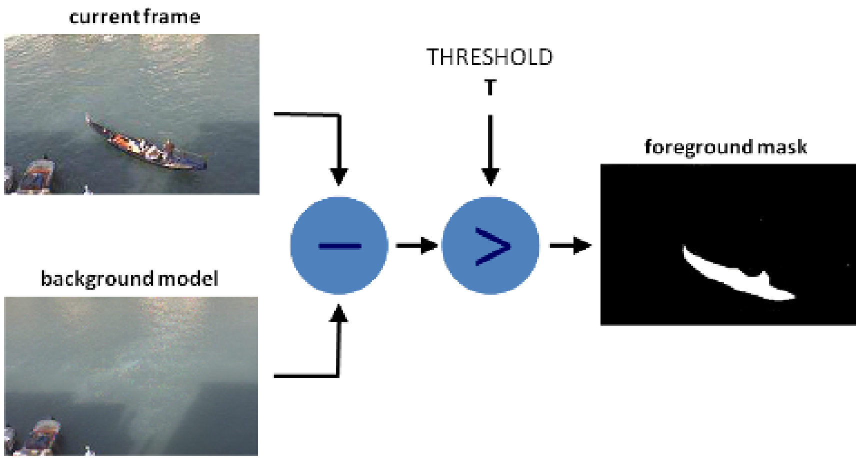
IMCV - FEUP

Input sequences characteristics

- Dataset: 2012
- Category: baseline
- Name: pedestrians
- Frames: 1099 (movement starts at frame 300)
- Dimensions: (240, 360, 3)

* Source: https://www.changedetection.net/

Background subtraction methods



Background Initialization:

- Use initial frames to create an average background image.

Background Subtraction:

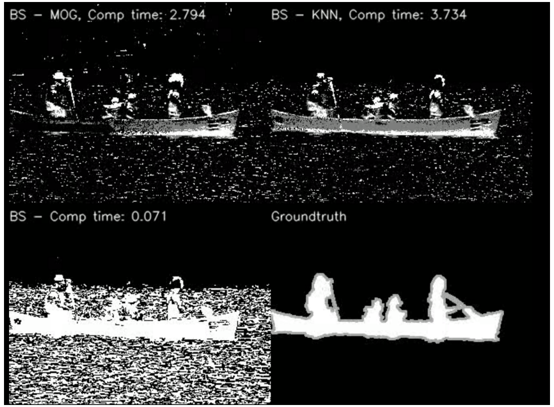
- Compare each subsequent frame to the established background.
- Identify significant changes using absolute differences and thresholding.

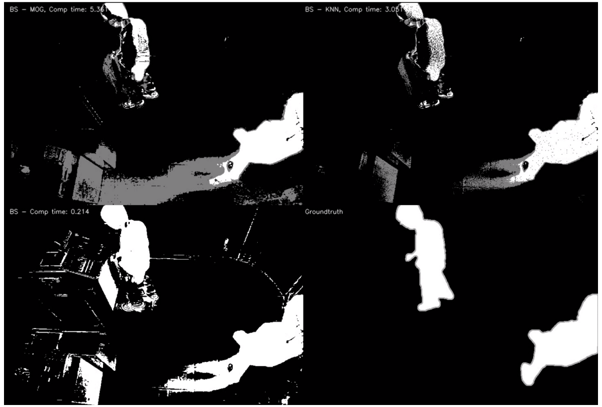
MOG (Mixture of Gaussians)
KNN (K-Nearest Neighbors)

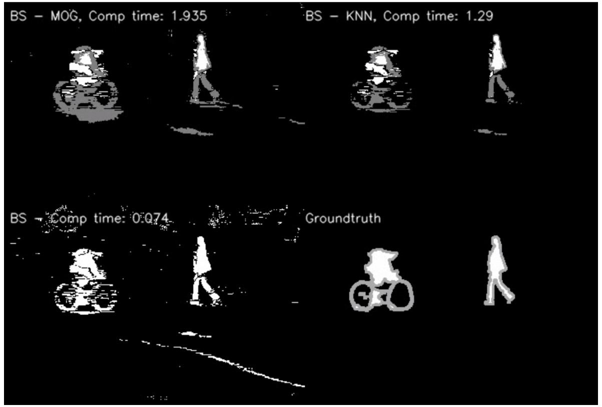
* Source: https://docs.opencv.org/4.x/d1/dc5/tutorial_background_subtraction.html

Background subtraction methods

- **MOG** - adapts to changes in the background model over time and is suitable for dynamic scenes where lighting conditions may change.
- **KNN** - considers the k-nearest neighbors' pixel values for each pixel, making it more adaptive to changes. It may be suitable for scenes with dynamic changes and adaptive to non-stationary backgrounds.
- **Custom "Mean" Method** - uses a simple mean of initial frames to estimate the background. Works well for static scenes where the background does not change significantly.







Frame differencing - Method

Differencing and summing technique (DST) is used for identifying moving objects and the objects are segmented from the background using bonding box based on the maximum and minimum coordinates of the moving objects.

Background Initialization:

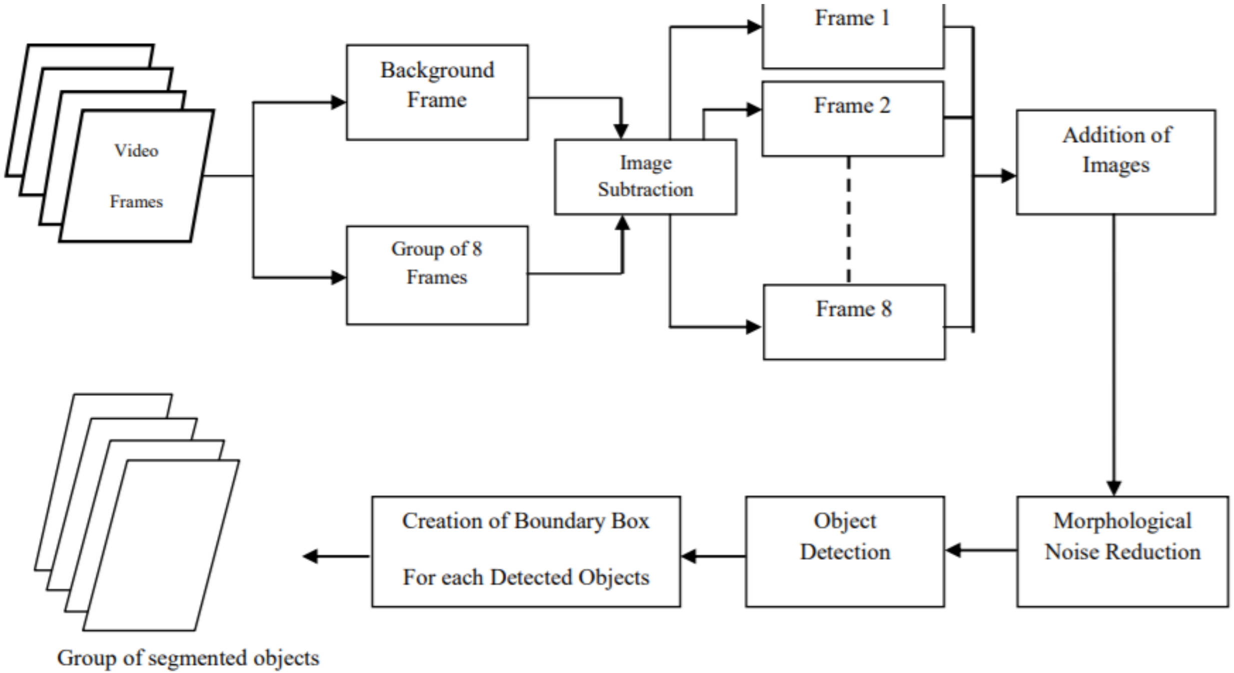
- Use initial frames to create an median background image.

Background Subtraction:

- Eight consecutive frames are considered and subtracted from the background.
- Difference images are added together to obtain a resultant image.
- The final image represents the max movement of the objects.

Frame differencing - Method

Block diagram for the segmentation of moving objects from the frames



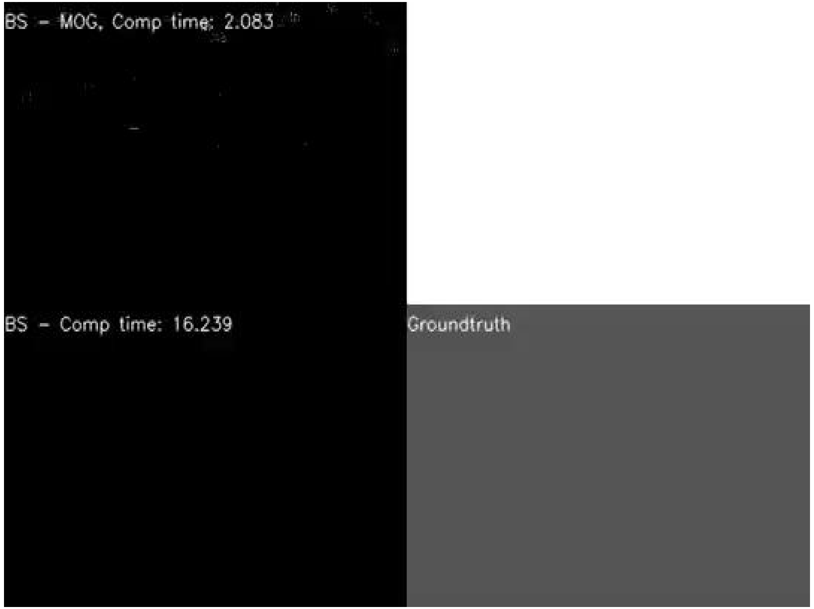
* Source: https://research.ijcaonline.org/volume102/number7/pxc3898647.pdf

Results: Metrics

	Efficiency (Seconds)	Precision	Accuracy	DICE	IoU
Background subtraction MOG	2.083	0.621	0.991	0.617	0.521
Background subtraction KNN	2.562	0.828	0.993	0.716	0.622
Background subtraction Mean	16.239	0.343	0.967	0.413	0.289
Frame differencing	1.023	0.295	0.505	0.588	0.503

Results: Output sequences

Background Subtraction



Frame differencing

Thank you!